# Erik H. Kramer

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## Education

**University of California, Los Angeles (UCLA)** - Ph.D. and M.S. in Mechanical Engineering

- Dissertation: *Robotic Physical Emulation of Virtual Dynamic Environments: Applications in Full Body Haptics and Spacecraft Development*

**University of California, Berkeley** - B.A. in Physics and B.S. in Mechanical Engineering

## Robotics Hardware & Mechanical Systems Experience

**NASA Jet Propulsion Laboratory** - Mechanical Engineering Postdoctoral Fellow *Mar 2025 - Present*

- Responsible engineer for a mobile manipulation robotic testbed, owning development and maintenance of electromechanical and software systems
- Led technical execution for TRL-4/5 autonomous system demos of flight-like satellite servicing, rover payload transfer, and telescope assembly
- Served as Deputy Task Manager for a 6-person lunar solar array assembly technology development team, leading technical efforts to meet milestones
- Leading design of updated rover steering actuators, using load-case analysis to guide development while matching existing hardware interfaces
- Diagnosed, repaired, and documented root causes of actuator failures through hardware teardown and reassembly, restoring system functionality
- Operated multiple robotic arm platforms for test campaigns, rapidly resolving electromechanical and controls issues in real time to maintain progress
- Developed driving algorithms for a robotic mobility system and analyzed field-test data to validate algorithm functionality and actuator performance
- Designed robotic sampling sequences to show lunar scoop collection of bulk regolith and rocks over repeated passes for a mission concept review

**Mechanical Engineering Intern: Advanced Robotic Systems Group** *Mar 2024 - Dec 2024*

- Developed and delivered the central structure for a hazardous environment rover, performing design, analysis, procurement, and integration
- Designed a new robotic arm optimized for sampling kinematics, meeting torque and stiffness requirements while leveraging heritage hardware
- Drove full-cycle development of a robotic in-space servicing, assembly, and manufacturing testbed, from conceptual design through initial bring-up
- Applied rapid design techniques to quickly develop test rigs and robot hardware components while supporting rover field testing campaigns

**Robotics Intern: Robotic Mobility Group** *Mar 2022 - Mar 2024*

- Architected features for a robot arm & lander testbed using C++, such as a torque control algorithm and a ROS interface for an autonomy subsystem
- Optimized motion planning for a redundant 7-DOF robotic arm using analytic inverse kinematics, increasing algorithm speed by 30%
- Analyzed large datasets of robot telemetry to understand the root cause of unexpected hardware faults during operation and implement code fixes
- Received Notable Organizational Value Added Award for outstanding development of a low-gravity dynamics motion controller for a physical testbed

**UCLA Bionics Lab** - Graduate Research Engineer *Sept 2015 - Dec 2024*

- Led a team in the end-to-end development of a robotic exoskeleton, providing hands-on work for design, fabrication, assembly, testing, and software
- Integrated force-torque sensors with 5 industrial robotic arms by developing real-time C++ software, resulting in safe multi-robot coordinated control
- Built MuJoCo dynamics simulations and admittance control algorithms, enabling safe human-robot haptic interaction with virtual environments
- Translated high-level requirements into hardware configurations through kinematic workspace analysis, achieving >96% coverage of potential users
- Collaborated with analysts to perform hand calculations and conduct stress FEA, ensuring safety factors of structural weldments met requirements
- Utilized CAD to design small connectors and large structures for manufacturability with traditional processes such as CNC and welding

## Additional Engineering & Science Experience

**Physics & Astronomy Department at UCLA** - Lead Teaching Fellow *Sept 2016 - Mar 2022*

- Managed multiple small teams teaching space science, circuit design, microcontrollers, E&M, Python, data analysis, technical writing, and mechanics

**Super Cryogenic Dark Matter Search Collaboration** - Undergraduate Researcher *Feb 2013 - June 2015*

- Designed metallic and composite cryogenic instrument thermal standoffs and drafted part drawings using ASME Y14.5 GD&T standards
- Conducted stress and thermal simulations via FEA and custom MATLAB scripts on 3 potential designs to support trade studies
- Delivered mechanical test support equipment and used it to qualify standoff designs through strength and cryogenic thermal V&V testing

**Berkeley Particle Cosmology Group** - Undergraduate Researcher *Feb 2013 - June 2015*

- Translated rough ideas from scientific experts into functional mechanical hardware, such as cold plates, for sub-kelvin high-vacuum environments
- Assembled hardware in a cleanroom and maintained low-temp experiments in bath cryostat vacuum chambers and a dilution refrigerator
- Created procedures and documented build and test instructions for experiments determining low-temp properties of engineering materials

## Independent Projects & Mentorship

**FIRST Robotics Mentor** - Team 1138 *Jan 2026 - Present*

- Guiding high school students through mechanical design, electrical integration, software development, and design reviews for competition robots

**Costume & Mechatronic Prop Design** *July 2006 - Present*

- Built replicas and wearables, including a Mars 2020 RC rover and an animatronic tail by utilizing mechanical, electrical, and software design processes

## Skills

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| **Technical** | Mechanical design, Systems integration, GD&T, FEA, Test design, V&V, Failure analysis, Data analysis, Rapid prototyping, Robot control |
| **Hardware** | Robotic arms, Mobility systems, 3D Printers, Sensors, Material test & qualification systems, Soldering, Wire bonder, Machine shop tools |
| **Software** | SolidWorks, Teamcenter, Linux, Git, ROS, Bluehill Universal, Vicon Tracker, Slicing software, Photoshop, Excel, Google Workspace |
| **Coding** | C/C++, MATLAB, Python, LaTeX |